

KADINA, Australia

WMO: 946850

Lat: 33.9703S

Lon: 137.6628E

Elev: 42

StdP: 100.83

Time Zone: 9.50 (AUA)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.6	1.8	-3.4	2.8	25.0	-1.6	3.3	19.1	13.7	14.2	12.7	14.0	3.5	120	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.9	39.3	19.2	36.5	18.6	34.2	18.0	21.6	29.8	20.7	30.1	19.9	29.2	6.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.6	23.7	18.4	13.4	22.8	17.1	12.3	22.2	63.3	29.4	60.0	30.2	57.1	29.5	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	11.0	10.0	-1.4	43.2	0.7	1.6	-1.9	44.3	-2.3	45.3	-2.6	46.2	-3.1	47.3		
(5)			-1.4	23.0	1.0	0.9	-2.2	23.7	-2.8	24.2	-3.3	24.7	-4.1	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	23.4	22.5	20.5	17.1	13.7	10.8	10.2	10.5	12.7	15.7	18.7	20.8		
	DBStd	5.72	4.17	3.96	3.82	3.05	2.36	2.20	2.19	2.49	2.95	3.82	4.18	4.36		
	HDD10.0	71	0	0	0	0	1	17	25	23	5	1	0	0		
	HDD18.3	1289	3	5	17	59	144	227	252	242	172	102	46	20		
	CDD10.0	2391	416	351	327	214	116	41	31	39	85	177	261	335		
	CDD18.3	567	159	122	86	23	1	0	0	0	2	21	57	97		
	CDH23.3	7682	2091	1501	1021	285	9	0	0	3	60	402	879	1430		
	CDH26.7	3925	1172	787	488	92	1	0	0	0	11	156	435	784		
(14) Wind	WSAvg	5.4	6.0	5.8	5.4	5.0	4.8	4.6	5.1	5.2	5.2	5.6	5.7	6.0		
(15) Precipitation	PrecAvg	376	15	21	17	27	40	49	49	43	42	32	23	17		
	PrecMax	580	72	98	58	117	114	138	107	88	106	99	50	53		
	PrecMin	237	1	0	0	0	0	3	7	5	7	1	0	0		
	PrecStd	79	15	25	16	27	24	28	23	21	23	26	13	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	40.6	38.7	32.5	24.5	18.7	18.9	23.0	28.3	34.1	38.2	41.3		
		MCWB	20.6	20.4	18.2	17.3	15.0	12.3	12.6	12.7	14.5	16.1	17.8	19.2		
	2%	DB	40.1	37.0	34.4	29.2	22.2	17.2	16.7	20.0	24.9	30.9	34.5	37.7		
		MCWB	19.7	19.5	18.4	15.9	14.6	12.9	12.0	12.1	13.8	14.9	17.1	18.5		
	5%	DB	36.3	34.6	31.9	26.6	20.3	16.2	15.6	17.9	22.4	28.0	31.9	34.5		
		MCWB	19.1	18.7	18.6	15.3	14.2	12.7	11.8	11.9	13.1	14.7	16.4	17.5		
	10%	DB	33.0	31.6	29.1	24.1	18.7	15.2	14.7	16.0	19.9	25.1	28.6	31.2		
		MCWB	18.2	18.4	17.7	15.2	14.0	12.1	11.5	11.7	12.6	14.1	15.7	16.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	23.0	22.1	19.7	17.6	15.4	14.4	14.8	16.7	18.3	20.0	21.5		
		MCDB	28.6	29.9	28.6	23.7	20.7	16.4	16.4	18.2	23.3	25.4	29.3	29.4		
	2%	WB	21.6	21.5	20.8	18.2	16.2	14.1	13.2	13.6	15.1	17.0	18.8	20.2		
		MCDB	31.4	29.3	27.7	23.5	19.3	15.9	15.3	17.5	20.8	24.9	28.6	29.9		
	5%	WB	20.7	20.4	19.7	17.1	15.4	13.2	12.6	12.8	14.1	16.0	17.9	19.2		
		MCDB	31.7	29.8	27.2	23.2	18.5	15.2	14.7	16.1	19.7	23.8	27.2	28.9		
	10%	WB	19.7	19.5	18.8	16.3	14.5	12.5	11.8	12.1	13.3	15.1	16.9	18.1		
		MCDB	30.2	28.7	26.8	22.1	17.7	14.6	14.1	15.3	18.5	22.4	25.6	27.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	14.0	13.1	12.4	9.9	9.1	9.1	11.3	13.2	14.7	15.0	14.8		
		MCDBR	20.8	18.9	17.5	16.7	12.7	10.0	10.1	14.7	18.3	20.4	20.6	21.0		
		MCWBR	6.4	5.6	5.9	6.9	6.7	6.6	6.5	8.1	8.4	8.0	6.9	6.6		
	5% WB	MCDBR	15.0	14.3	13.7	11.8	9.5	7.9	8.4	11.4	14.4	15.2	15.6	15.3		
		MCWBR	6.2	5.5	5.3	6.1	5.8	5.6	6.0	7.3	7.7	6.9	6.1	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.344	0.326	0.318	0.308	0.297	0.299	0.306	0.333	0.342	0.346	0.338		
	taud		2.516	2.545	2.583	2.591	2.604	2.619	2.605	2.566	2.479	2.457	2.478	2.527		
	Ebn at Noon		991	972	955	907	864	849	866	907	931	962	984	1003		
	Edh at Noon		112	105	94	83	73	67	71	84	103	113	116	112		
(44) All-Sky Solar Radiation	RadAvg		7.89	7.00	5.72	4.24	3.00	2.48	2.70	3.66	4.97	6.30	7.31	7.78		
	RadStd		0.38	0.35	0.25	0.21	0.12	0.13	0.12	0.21	0.34	0.34	0.33	0.31		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.42	N/A	N/A	N/A	N/A	N/A	-92	+157	+69	

Nomenclature: See separate page